

Lesson 2: Arithmetic Sequences & Series — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Write down the first four terms of the arithmetic sequence with first term 4 and common difference 5.
2. Find the common difference of 11, 8, 5, 2, ...
3. An arithmetic sequence has $u_1 = 6$ and $d = 4$. Find u_7 .
4. Find the 12th term of 3, 10, 17, ...
5. An AP has $u_1 = 50$ and $d = -4$. Find u_{15} .
6. Find u_{20} for the AP with $u_1 = 8$, $d = 6$.

Section 2 — Core practice

7. Find the sum of the first 10 terms of the AP with $u_1 = 5$, $d = 4$.
8. Find S_{15} for the AP 2, 5, 8, ...
9. An AP has $u_1 = 100$, $d = -5$. Find the sum of the first 12 terms.
10. In an AP the 3rd term is 17 and the 7th term is 33. Find d and u_1 .
11. An AP has $u_2 = 9$ and $u_6 = 29$. Find u_1 and d .

Section 3 — Challenge & exam-style

- 12.** Which term of the AP 7, 12, 17, ... is equal to 82?
- 13.** A graduate's salary starts at £12000 and rises by £1500 each year. What is the salary in year 6?
- 14.** A theatre has 30 seats in row 1 and 4 more in each successive row. How many seats are in the first 18 rows in total?
- 15.** For the AP with $u_1 = 5$, $d = 3$, the sum of the first n terms is 124. Find n .